

Homework 2

Received Power and Link Geometry

Assigned: September 14, 2026 Due: September 28, 2026 Value: 100 points

Show your calculations and units. A calculator and the lecture slides are all you need. You may revise and resubmit your work until it is correct.

1. Slant Range • 20 Points

A spacecraft is in a circular orbit 500 km above Earth. Calculate its slant range from a ground station when its elevation angle is 10° . Use an Earth radius of 6,378 km, neglect station altitude, and round to the nearest kilometre. Set your calculator to degrees.

2. Voyager 2 Received Power • 60 Points

Calculate total received signal power for Voyager 2's X-band link to DSS-43, the 70 m antenna in Canberra. Use the updated range below with radio parameters from JPL's January 1, 1996 budget [1].

Range: 21,476,670,660 km; frequency: 8.415 GHz; transmit power: 12.3 W; transmit antenna gain: 48.20 dBi; receive antenna gain: 74.01 dBi. Use 0 dB transmit feed loss and 0.42 dB other losses (pointing, atmosphere, and polarization combined).

- Calculate [EIRP] in dBW. [20 points]
- Calculate free-space path loss [L_p] in dB. [20 points]
- Calculate received power [P_r] in dBW. Round your results to one decimal place. [20 points]

3. The Elevation Mask Trade • 20 Points

For the Earth-orbiting spacecraft in Problem 1, suppose we increase the minimum elevation angle used for contacts. In two or three sentences, explain what happens to the maximum useful slant range, the worst-case free-space path loss, and the available contact time. Keep the orbit, station, frequency, and antenna gains unchanged.

Reference Formulas

$$d = \sqrt{(R_E + h)^2 - R_E^2 \cos^2 e} - R_E \sin e$$

$$[P] = 10 \log_{10}(P / 1 \text{ W})$$

$$[EIRP] = [P_t] - [L_t] + [G_t]$$

$$[L_p] = 92.45 + 20 \log_{10}(R) + 20 \log_{10}(f)$$

$$[P_r] = [EIRP] - [L_p] + [G_r] - [L_{other}]$$

For slant range, h is altitude and e is elevation. In the path-loss formula, enter R in km and f in GHz. Brackets denote decibel values. Power is in dBW, antenna gains are in dBi, and losses are in dB. Record losses as positive amounts and subtract them.

[1] Ludwig and Taylor, Voyager Telecommunications, JPL DESCANSO Article 4 (2002), Table 5-3, p. 26. Power converted from 40.90 dBm and rounded to 12.3 W. Other losses sum to 0.42 dB. Range updated from the supplied Voyager 2 distance display on September 15, 2026 (rounded to the nearest km).

JPL source: Voyager Telecommunications